Anton Kelly Robert P. Estac

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WORK EXPERIENCE

San Pablo Water District **Feb. 3 – April. 1**
Internship *San Pablo City, Laguna*

- Assisted in technical work around the whole water district office.
- Assisted test running on new systems and create a documentation on how the system works.

EDUCATION

Laguna State Polytechnic University – San Pablo City Campus **June, 2026**
BS Computer Engineering *San Pablo City, Laguna*

- GWA: 1.83

ACADEMIC PROJECTS

- **Thesis Project: Game-Based Learning: An Arcade System for Improving Sign Language Skills of Children with Hearing and Speech Impairments** | Lead Game Developer and Designer
  - Designed and developed a 2D educational game, engineering intuitive levels and gameplay mechanics based around Filipino Sign Language (FSL) learning.
  - Implemented a low-latency UDP communication pipeline using Python to connect the FSL recognition model to the game's core input loop for real-time responsiveness.
  - Conducted comprehensive functionality, efficiency, accuracy, reliability, and effectiveness testing matrices to validate system-wide hardware and software performance benchmarks.
- **AI-Powered Elderly Health Risk with Website Dashboard** | *Embedded/Backend Developer*
  - Programmed a backend API system that handles data routing and user authentication using FastAPI and Python.
  - Designed and deployed a normalized MySQL database schema optimized for the efficient storage and retrieval of continuous health monitoring metrics.
  - Managed the end-to-end deployment pipeline, configuring and hosting the backend services and database infrastructure on cloud platforms (Railway/Vercel) for continuous availability.
- **ESP32 Cam Object Detection** | AI/ML Developer
  - Developed and deployed a custom machine learning model via Edge Impulse to an ESP32-Cam for real-time electronic component detection (e.g., batteries, sensors, breadboards).
  - Built an ML-powered hardware recognition system by training a custom model on Edge Impulse and flashing it directly to an ESP32-Cam microcontroller.

SKILLS

Communication; Adaptability and Proactive; Teamwork and Work Ethics; AutoCAD; KiCad; SketchUp; GitHub; Unity; Arduino; Microcontrollers; Hardware troubleshooting; System integration; Python; GoLang; C#; MS Office (Word, Excel, PowerPoint)